

Análisis de expresión diferencial - clase

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Librerías

```
# install.packages("BiocManager")

# install.packages("tidyverse")
# install.packages("ggplot2")
# BiocManager::install("edgeR")
# BiocManager::install("clusterProfiler")
# BiocManager::install("msigdb")

library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2     4.0.1      v tibble     3.3.1
## v lubridate  1.9.4      v tidyr      1.3.2
## v purrr       1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(ggplot2)
library(edgeR)

## Loading required package: limma

library(clusterProfiler)

##
## clusterProfiler v4.16.0 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##
## Please cite:
##
## Guangchuang Yu, Li-Gen Wang, Yanyan Han and Qing-Yu He.
## clusterProfiler: an R package for comparing biological themes among
## gene clusters. OMICS: A Journal of Integrative Biology. 2012,
```

```
## 16(5):284-287
##
## Attaching package: 'clusterProfiler'
##
## The following object is masked from 'package:purrr':
##
##     simplify
##
## The following object is masked from 'package:stats':
##
##     filter
```

```
library(msigdb)

packageVersion("tidyverse")
```

```
## [1] '2.0.0'
```

```
packageVersion("ggplot2")
```

```
## [1] '4.0.1'
```

```
packageVersion("edgeR")
```

```
## [1] '4.6.3'
```

```
packageVersion("clusterProfiler")
```

```
## [1] '4.16.0'
```

```
packageVersion("msigdb")
```

```
## [1] '25.1.1'
```

Cargar datos

```
rna <- read.table(file="../input/coad_rna.tsv", sep="\t", header=TRUE, check.names = FALSE, row.names =
metadata <- read.table(file="../input/coad_metadata.tsv", sep="\t", header = TRUE)

head(metadata)
```

```
##      PATIENT_ID      SAMPLE_ID SUBTYPE AGE  SEX OS_STATUS
## 1 TCGA-A6-5667 TCGA-A6-5667-01    CIN  40 Female  0:LIVING
## 2 TCGA-AY-4070 TCGA-AY-4070-01    CIN  50 Female 1:DECEASED
## 3 TCGA-CM-6168 TCGA-CM-6168-01     GS  84 Female  0:LIVING
## 4 TCGA-CA-6716 TCGA-CA-6716-01    CIN  65   Male  0:LIVING
## 5 TCGA-F4-6569 TCGA-F4-6569-01    CIN  60   Male  0:LIVING
```

```
## 6 TCGA-A6-A56B TCGA-A6-A56B-01 CIN 57 Male 1:DECEASED
## COUNTRY_OF_RESIDENCE SITE_OF_BIOPSY purity ploidy wholeGenomeDuplication
## 1 <NA> Sigmoid colon 0.69 3.20 true
## 2 <NA> Cecum 0.31 4.39 true
## 3 United States Ascending colon 0.35 2.17 false
## 4 Vietnam Ascending colon 0.73 3.74 true
## 5 Russia Transverse colon 0.40 2.16 false
## 6 United States Sigmoid colon 0.72 3.21 true
## CMS metastasis
## 1 CMS2 false
## 2 no_label false
## 3 CMS4 false
## 4 CMS2 false
## 5 CMS4 false
## 6 no_label false
```

```
dim(rna)
```

```
## [1] 60660 300
```

Extraer el dato a comparar

El modelo edgeR necesita una colData que tenga en las filas las muestras, y la columna con el valor de

```
coldata <- metadata %>% select(PATIENT_ID, wholeGenomeDuplication) %>%
  column_to_rownames(var = "PATIENT_ID") %>%
  mutate(wholeGenomeDuplication = as.factor(wholeGenomeDuplication))
```

```
table(coldata)
```

```
## wholeGenomeDuplication
## false true
## 178 122
```

eliminar genes sin expresion

```
table(rowSums(rna) > 0) # TRUE genes con expresion, FALSE genes donde en todas las muestras son 0
```

```
##
## FALSE TRUE
## 12252 48408
```

```
rna <- rna[rowSums(rna) > 0, ]
dim(rna)
```

```
## [1] 48408 300
```

Correr análisis de expresión

```
# Alinear las muestras y usar false como referencia: el contraste sera true vs false
stopifnot(setequal(colnames(rna), rownames(coldata)))
coldata <- coldata[colnames(rna), , drop = FALSE]
coldata$wholeGenomeDuplication <- releve(coldata$wholeGenomeDuplication, ref = "false")
design <- model.matrix(~wholeGenomeDuplication, data = coldata)
```

```
dge <- DGEList(counts = rna)
# Filtrar genes con expresion insuficiente para el diseño y normalizar por TMM
keep <- filterByExpr(dge, design = design)
dge <- dge[keep, , keep.lib.sizes = FALSE]
dge <- calcNormFactors(dge, method = "TMM")
```

```
dge <- estimateDisp(dge, design)
fit <- glmQLFit(dge, design)
qlf <- glmQLFTest(fit, coef = "wholeGenomeDuplicationtrue")
colnames(design)
```

```
## [1] "(Intercept)" "wholeGenomeDuplicationtrue"
```

```
res <- topTags(qlf, n = Inf, sort.by = "none")$table %>%
  rename(log2FoldChange = logFC, padj = FDR)
summary(res)
```

```
## log2FoldChange      logCPM          F          PValue
## Min.      :-3.045368   Min.      : 3.310   Min.      : 0.0000   Min.      :0.000000
## 1st Qu.: -0.107840   1st Qu.: 4.210   1st Qu.: 0.5266   1st Qu.:0.009969
## Median : -0.005477   Median : 4.992   Median : 2.3488   Median :0.126425
## Mean    : -0.007964   Mean      : 5.293   Mean      : 5.1124   Mean     :0.264995
## 3rd Qu.:  0.104177   3rd Qu.: 6.035   3rd Qu.: 6.7252   3rd Qu.:0.468591
## Max.     :  1.618942   Max.      :15.094   Max.      :74.9484   Max.     :0.999846
##      padj
## Min.      :0.00000
## 1st Qu.:0.03986
## Median :0.25282
## Mean      :0.34880
## 3rd Qu.:0.62477
## Max.      :0.99985
```

```
# log2FC > 0 - enriquecido en true, < 0 enriquecido en false
# Un log2FC de 1 indica un cambio de dos veces. Es decir, si es 1 entonces en esa condicion esta expres
true_enriched <- res %>% filter(log2FoldChange > 1, padj < 0.1)
false_enriched <- res %>% filter(log2FoldChange < -1, padj < 0.1)

cat("Hay", nrow(true_enriched), "genes sobreexpresados en duplicacion de genoma completo, versus", nrow
```

```
## Hay 8 genes sobreexpresados en duplicacion de genoma completo, versus 35 subexpresados.
```

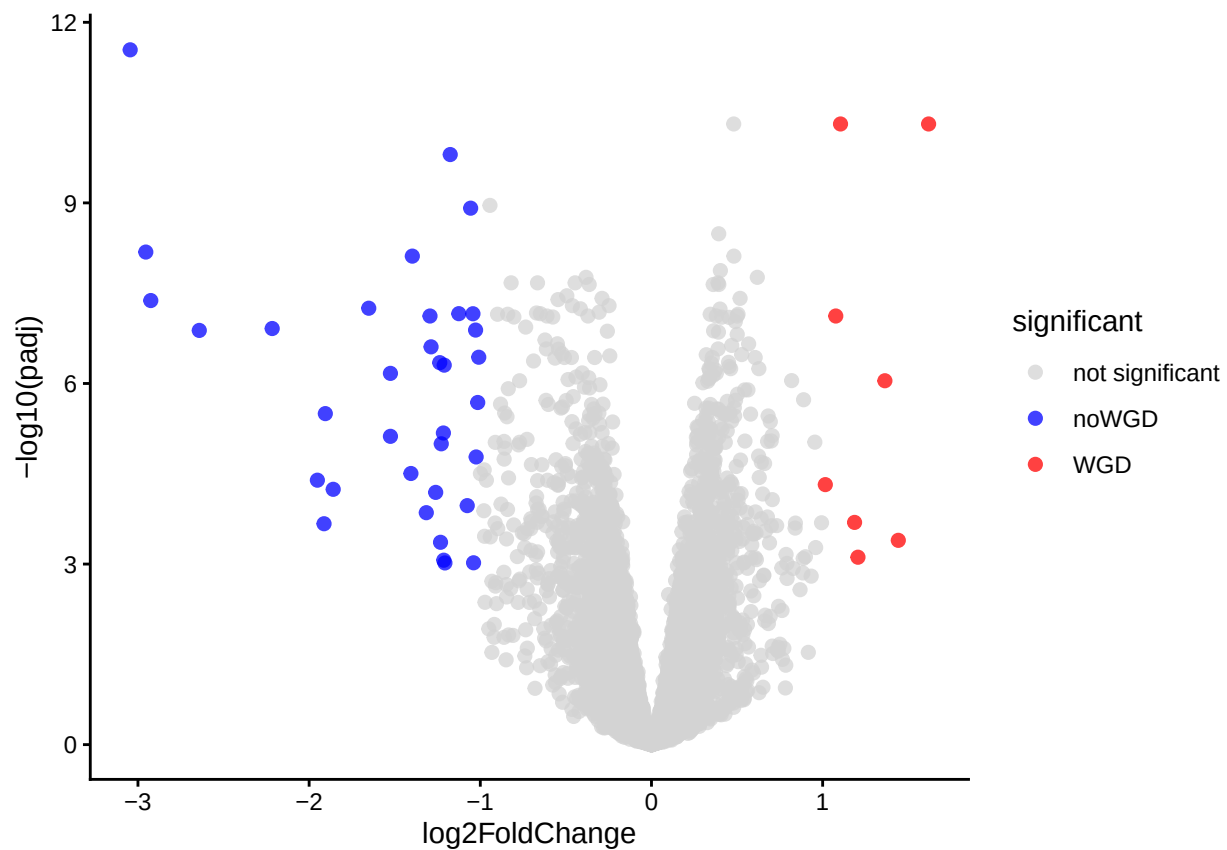
Volcanoplot

```

# Definir colores
res <- res %>%
  mutate(gene = rownames(res),
         significant = ifelse(gene %in% rownames(true_enriched), "WGD",
                             ifelse(gene %in% rownames(false_enriched),
                                    "noWGD",
                                    "not significant"))))

# generar volcano
ggplot(res, aes(x = log2FoldChange, y = -log10(padj),
               color = significant)) +
  geom_point(alpha = 0.75, size = 2) +
  scale_color_manual(values = c("WGD" = "red",
                              "noWGD" = "blue",
                              "not significant" = "lightgray")) +
  theme_classic()

```



```

# Definir una base de datos. Por ejemplo = GO procesos biológicos
go_bp <- msigdb(species = "Homo sapiens",
                collection = "C5",
                subcollection = "GO:BP")

# GSEA utiliza todos los genes evaluados, sin filtrar por log2FC o padj.
# La raíz del estadístico F con el signo del log2FC ordena true vs false.
gene_list <- setNames(sign(res$log2FoldChange) * sqrt(res$F), res$gene)

```

```
gene_list <- sort(gene_list[is.finite(gene_list)], decreasing = TRUE)

# Se establece una tabla de asociacion proceso-gen
term2gene <- unique(as.data.frame(go_bp[, c("gs_name", "ensembl_gene"))))
```

```
# NES > 0: enriquecido en WGD; NES < 0: enriquecido en noWGD
set.seed(123)
enrich <- GSEA(geneList = gene_list,
               TERM2GENE = term2gene,
               pAdjustMethod = "BH",
               pvalueCutoff = 0.1,
               minGSSize = 3,
               eps = 0,
               nPermSimple = 10000,
               seed = TRUE,
               by = "fgsea",
               BPPARAM = BiocParallel::SerialParam())
```

```
## using 'fgsea' for GSEA analysis, please cite Korotkevich et al (2019).
```

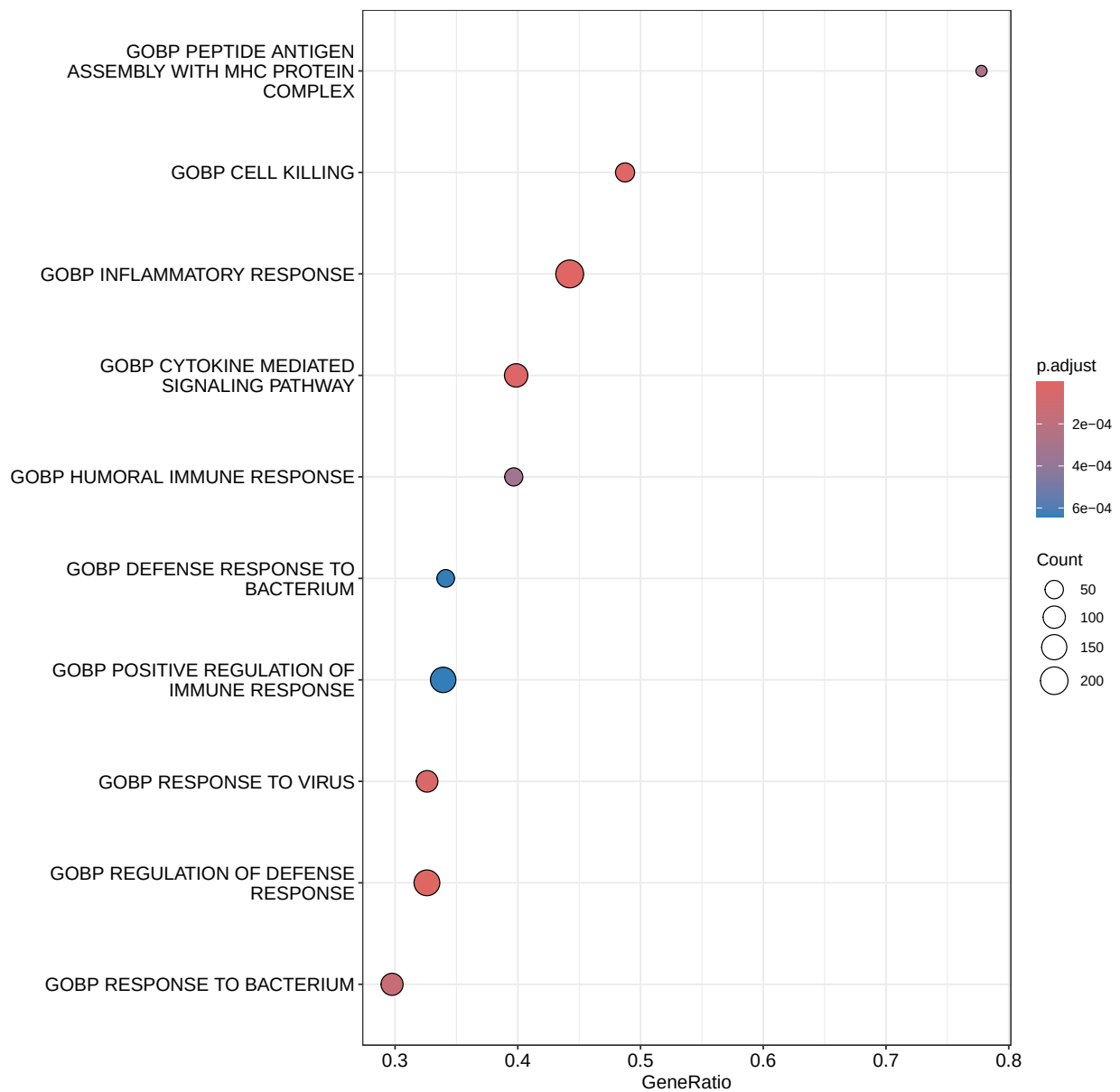
```
## preparing geneSet collections...
```

```
## GSEA analysis...
```

```
## leading edge analysis...
```

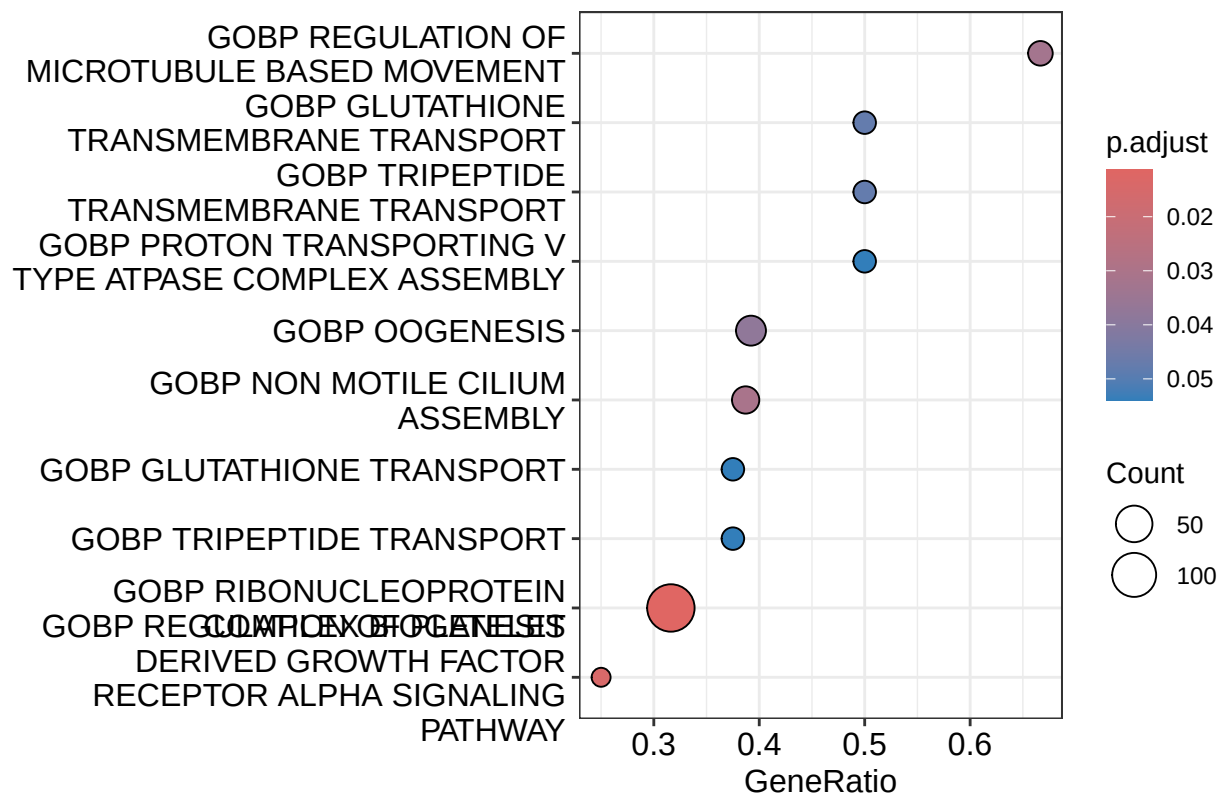
```
## done...
```

```
dotplot(enrich, showCategory = 10,
        color = "p.adjust")
```

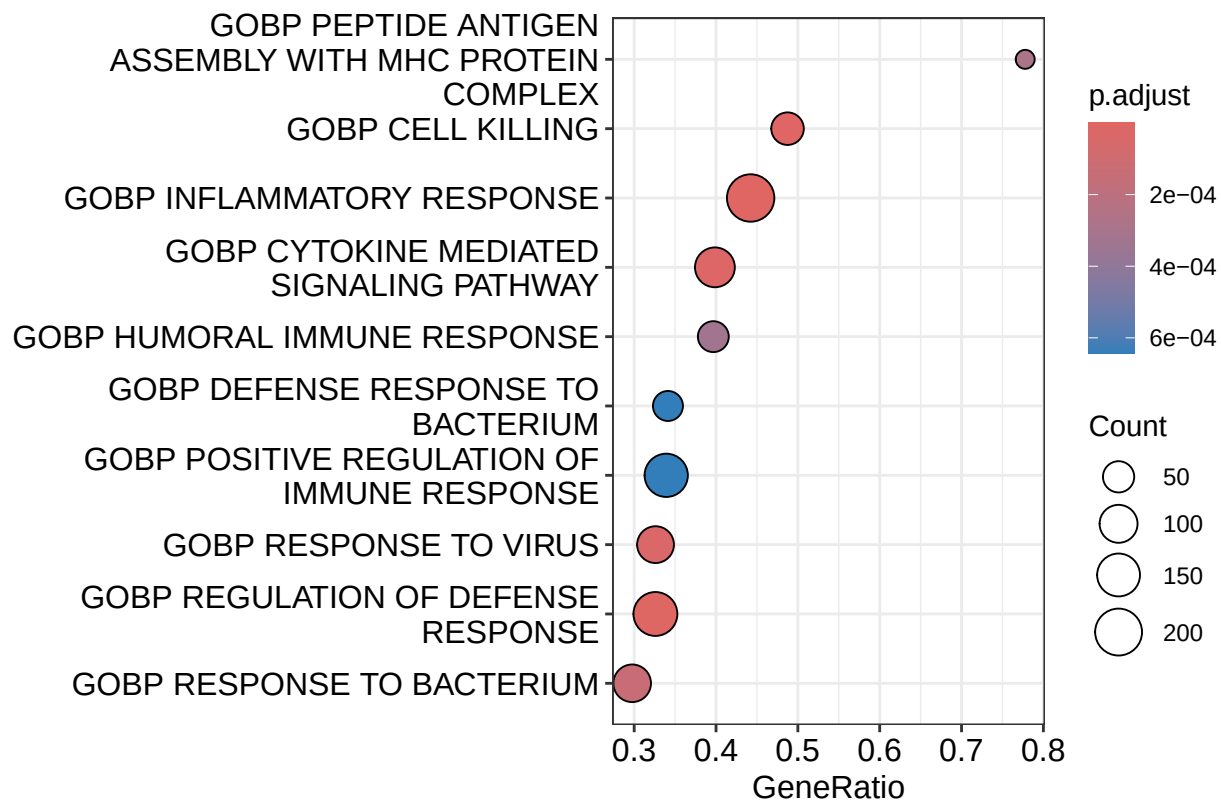


Alternativa: visualizar el enriquecimiento por categoria (segun el signo del NES)

```
wgd_enrich <- enrich
wgd_enrich@result <- enrich@result[enrich@result$NES > 0, , drop = FALSE]
dotplot(wgd_enrich, showCategory = 10,
        color = "p.adjust")
```



```
nowgd_enrich <- enrich
nowgd_enrich@result <- enrich@result[enrich@result$NES < 0, , drop = FALSE]
dotplot(nowgd_enrich, showCategory = 10,
        color = "p.adjust")
```



```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13)
## Platform: x86_64-pc-linux-gnu
## Running under: AlmaLinux 9.5 (Teal Serval)
##
## Matrix products: default
## BLAS: /mnt/beegfs/apps/r-4.5.1/lib64/R/lib/libRblas.so
## LAPACK: /mnt/beegfs/apps/r-4.5.1/lib64/R/lib/libRlapack.so; LAPACK version 3.12.1
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8 LC_NUMERIC=C
## [3] LC_TIME=en_US.UTF-8 LC_COLLATE=en_US.UTF-8
## [5] LC_MONETARY=en_US.UTF-8 LC_MESSAGES=en_US.UTF-8
## [7] LC_PAPER=en_US.UTF-8 LC_NAME=C
## [9] LC_ADDRESS=C LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: US/Eastern
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
## [1] msigdb_r25.1.1 clusterProfiler_4.16.0 edgeR_4.6.3
```

```

## [4] limma_3.64.3          lubridate_1.9.4      forcats_1.0.1
## [7] stringr_1.6.0         dplyr_1.1.4         purrr_1.2.1
## [10] readr_2.1.6           tidyr_1.3.2         tibble_3.3.1
## [13] ggplot2_4.0.1         tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] DBI_1.2.3              gson_0.1.0           rlang_1.1.7
## [4] magrittr_2.0.4         DOSE_4.2.0           otel_0.2.0
## [7] compiler_4.5.1        RSQLite_2.4.5        png_0.1-8
## [10] vctrs_0.7.0           reshape2_1.4.5       pkgconfig_2.0.3
## [13] crayon_1.5.3          fastmap_1.2.0        XVector_0.48.0
## [16] labeling_0.4.3        rmarkdown_2.30       tzdb_0.5.0
## [19] enrichplot_1.28.4     UCSC.utils_1.4.0     bit_4.6.0
## [22] xfun_0.56             cachem_1.1.0         aplot_0.2.9
## [25] GenomeInfoDb_1.44.1   jsonlite_2.0.0       blob_1.3.0
## [28] BiocParallel_1.42.1   parallel_4.5.1       R6_2.6.1
## [31] stringi_1.8.7         RColorBrewer_1.1-3   GOSeqSim_2.34.0
## [34] assertthat_0.2.1      Rcpp_1.1.1           knitr_1.51
## [37] ggtangle_0.1.1        R.utils_2.13.0       IRanges_2.42.0
## [40] Matrix_1.7-3          splines_4.5.1        igraph_2.2.1
## [43] timechange_0.3.0      tidyselect_1.2.1     qvalue_2.40.0
## [46] rstudioapi_0.18.0     dichromat_2.0-0.1    yaml_2.3.12
## [49] codetools_0.2-20      curl_7.0.0           lattice_0.22-7
## [52] plyr_1.8.9            treeio_1.32.0        Biobase_2.68.0
## [55] withr_3.0.2           KEGGREST_1.48.1      S7_0.2.1
## [58] evaluate_1.0.5        gridGraphics_0.5-1   Biostrings_2.76.0
## [61] ggtree_3.16.3         pillar_1.11.1        stats4_4.5.1
## [64] ggfun_0.2.0           generics_0.1.4       S4Vectors_0.48.0
## [67] hms_1.1.4            tidytree_0.4.7       scales_1.4.0
## [70] glue_1.8.0           lazyeval_0.2.2       tools_4.5.1
## [73] data.table_1.18.0     fgsea_1.34.2         locfit_1.5-9.12
## [76] babelgene_22.9        fs_1.6.6             fastmatch_1.1-8
## [79] cowplot_1.2.0         grid_4.5.1           ape_5.8-1
## [82] AnnotationDbi_1.72.0  nlme_3.1-168         GenomeInfoDbData_1.2.14
## [85] patchwork_1.3.2       cli_3.6.5            rappdirs_0.3.4
## [88] gtable_0.3.6          R.methodsS3_1.8.2    yulab.utils_0.2.3
## [91] digest_0.6.39         BiocGenerics_0.54.0  ggrepel_0.9.6
## [94] ggplotify_0.1.3       farver_2.1.2         memoise_2.0.1
## [97] htmltools_0.5.9       R.oo_1.27.1          lifecycle_1.0.5
## [100] httr_1.4.7           GO.db_3.21.0         statmod_1.5.1
## [103] bit64_4.6.0-1

```